gpeyt1-11 · OSHA sloping

GEO PE · Long Calculation Problems I — Topic 3: Construction Observation, QA/QC & Safety

Problem

A 13 ft deep utility trench with a 5 ft wide bottom will be sloped (no shoring) on both sides through a medium-stiff silty clay classified by the competent person as OSHA Type B (qu = 1.0 tsf, fissured, previously disturbed). The trench run is 240 ft long. The spoil pile will be kept 2 ft back from the edge, and the excavator weighs 48,000 lb.

Most nearly, the minimum top width of the trench is:

A. 24 ft
B. 44 ft
C. 31 ft
D. 18 ft

Given equations / data

> OSHA maximum allowable slopes (H:V): Type A 3/4:1, Type B 1:1, Type C 1½:1 (excavations ≤ 20 ft)

Worked solution

1. Type B → 1H:1V on each side
2. Horizontal run each side = 13 × 1 = 13 ft
3. Top width = 5 + 2(13) = 31.0 ft
4. (Excavated volume ≈ (5+31.0)/2 × 13 × 240/27 = 2,080 yd³ — not asked)

Final answer

31 ft

Why the other values are wrong

- **24 ft** — used the Type A slope 3/4:1
- **44 ft** — used the Type C slope 1½:1
- **18 ft** — sloped only one side

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.10.2 Slope and Shield Configurations (Allowable Slopes table), p. 152; 29 CFR 1926 Subpart P App. B

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=hE7KZATybxA, https://www.youtube.com/watch?v=5IweGa4yOBc (no creator values copied).

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